Trigonometric Identities — Cheat Sheet

Basic / Definitions

Reciprocals: $\sec x = \dfrac{1}{\cos x}$, $\csc x = \dfrac{1}{\sin x}$, $\cot x = \dfrac{\cos x}{\sin x}$.
Quotients: $\tan x = \dfrac{\sin x}{\cos x}$, $\cot x = \dfrac{1}{\tan x}$.
Pythagorean: $\sin^2 x + \cos^2 x = 1$, $1 + \tan^2 x = \sec^2 x$, $1 + \cot^2 x = \csc^2 x$.
Negative angle: $\sin(-x) = -\sin x$, $\cos(-x) = \cos x$, $\tan(-x) = -\tan x$.
Periodicity: $\sin(x + 2\pi) = \sin x$, $\cos(x + 2\pi) = \cos x$, $\tan(x + \pi) = \tan x$.

Angle Addition / Subtraction

$$\begin{aligned}
\sin(x \pm y) &= \sin x \cos y \pm \cos x \sin y, \\
\cos(x \pm y) &= \cos x \cos y \mp \sin x \sin y, \\
\tan(x \pm y) &= \frac{\tan x \pm \tan y}{1 \mp \tan x \ \tan y}.
\end{aligned}$$

Double / Triple Angle

$$\begin{aligned}
\sin(2x) &= 2 \sin x \cos x, \\
\cos(2x) &= \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x, \\
\tan(2x) &= \frac{2\tan x}{1 - \tan^2 x} \quad (\tan x \neq \pm 1), \\
\sin(3x) &= 3\sin x - 4\sin^3 x, \qquad \cos(3x) = 4\cos^3 x - 3\cos x.
\end{aligned}$$

Half-Angle / Power-Reducing

$$\begin{aligned}
\sin^2 x &= \frac{1 - \cos(2x)}{2}, \qquad \cos^2 x = \frac{1 + \cos(2x)}{2}, \\
\sin^2 \frac{x}{2} &= \frac{1 - \cos x}{2}, \qquad \cos^2 \frac{x}{2} = \frac{1 + \cos x}{2}, \\
\tan \frac{x}{2} &= \frac{\sin x}{1 + \cos x} = \frac{1 - \cos x}{\sin x}.
\end{aligned}$$

Product-to-Sum & Sum-to-Product

$$\begin{aligned}
\sin x \cos y &= \tfrac{1}{2}\left[\sin(x+y) + \sin(x-y)\right], \\
\cos x \cos y &= \tfrac{1}{2}\left[\cos(x+y) + \cos(x-y)\right], \\
\sin x \sin y &= \tfrac{1}{2}\left[\cos(x-y) - \cos(x+y)\right],
\end{aligned}$$

$$\sin x + \sin y = 2\sin\frac{x+y}{2}\cos\frac{x-y}{2}, \qquad \sin x - \sin y = 2\cos\frac{x+y}{2}\sin\frac{x-y}{2},$$
$$\cos x + \cos y = 2\cos\frac{x+y}{2}\cos\frac{x-y}{2}, \qquad \cos x - \cos y = -2\sin\frac{x+y}{2}\sin\frac{x-y}{2}.$$

Useful for Integration

$$\int \sin^2 x\, dx = \frac{x}{2} - \frac{\sin(2x)}{4} + C, \qquad \int \cos^2 x\, dx = \frac{x}{2} + \frac{\sin(2x)}{4} + C,$$
$$\int \sin x \cos x\, dx = -\frac{\cos^2 x}{2} + C = \frac{\sin^2 x}{2} + C,$$
$$\sin^2 x\ \cos^2 x = \tfrac{1}{4}\sin^2(2x), \qquad \int \sin^2 x\ \cos^2 x\, dx = \frac{x}{8} - \frac{\sin(4x)}{32} + C.$$

Complex (Euler) Forms

$$e^{ix} = \cos x + i \sin x, \qquad \cos x = \tfrac{e^{ix}+e^{-ix}}{2}, \qquad \sin x = \tfrac{e^{ix}-e^{-ix}}{2i}.$$

Parity tip: cos is even, sin is odd. On $[-L, L]$, integrals of odd functions vanish; even functions double: $\int_{-L}^{L}$ even $= 2\int_0^L$.